

BCA(Semester-II) Examination, 2022

(Session : 2020-23)

COMPUTER APPLICATION

[Paper Code : BCA-203]

(Data Structure Through "C")

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answer in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is data structure? Explain the types of linear data structure with diagram.
2. What is function? Explain the types of function. Write a C program to find all prime number from 2 to n given number using function.
3. What is circular link list? Write a C program to demonstrate the concept of circular link
4. What is queue? Write a C program to demonstrate add delete operation in queue using array implementation.
5. What is Binary tree? Discuss its properties.
6. What is sorting? Discuss the types of sorting.
7. What is linear searching? Write a C program to enter n number in an array. Search a number in the given list of array using linear search.
8. What is an array? Discuss the types of array. write a C program to find sum of all even number from n given number.
9. Explain the concept of double linked list. Write a C program to perform the following operation in a double linked list:
 - (a) Create a node
 - (b) Insert a node
 - (c) Delete a node
 - (d) Searching of nodes
10. Write notes on:
 - (a) Graph
 - (b) Structure
 - (c) Tree Traversalslist.

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(d) Recursion Vs. Looping